

4

FLOWS ON THE CIRCLE

4.0 Introduction

So far we've concentrated on the equation $\dot{x} = f(x)$, which we visualized as a vector field on the line. Now it's time to consider a new kind of differential equation and its corresponding phase space. This equation,

$$\dot{\theta} = f(\theta),$$

corresponds to a **vector field on the circle**. Here θ is a point on the circle and $\dot{\theta}$ is the velocity vector at that point, determined by the rule $\dot{\theta} = f(\theta)$. Like the line, the circle is one-dimensional, but it has an important new property: by flowing in one direction, a particle can eventually return to its starting place (Figure 4.0.1). Thus periodic solutions become possible for the first time in this book! To put it another way, *vector fields on the circle provide the most basic model of systems that can oscillate*.



Figure 4.0.1

However, in all other respects, flows on the circle are similar to flows on the line, so this will be a short chapter. We will discuss the dynamics of some simple oscillators, and then show that these equations arise in a wide variety of applications. For example, the flashing of fireflies and the voltage oscillations of superconducting Josephson junctions have been modeled by the same equation, even though their oscillation frequencies differ by about ten orders of magnitude!

4.1 Examples and Definitions

Let's begin with some examples, and then give a more careful definition of vector fields on the circle.

EXAMPLE 4.1.1:

Sketch the vector field on the circle corresponding to $\dot{\theta} = \sin \theta$.

Solution: We assign coordinates to the circle in the usual way, with $\theta = 0$ in the direction of “east,” and with θ increasing counterclockwise.

To sketch the vector field, we first find the fixed points, defined by $\dot{\theta} = 0$. These occur at $\theta^* = 0$ and $\theta^* = \pi$. To determine their stability, note that $\sin \theta > 0$ on the upper semicircle. Hence $\dot{\theta} > 0$, so the flow is counterclockwise. Similarly, the flow is clockwise on the lower semicircle, where $\dot{\theta} < 0$. Hence $\theta^* = \pi$ is stable and $\theta^* = 0$ is unstable, as shown in Figure 4.1.1.

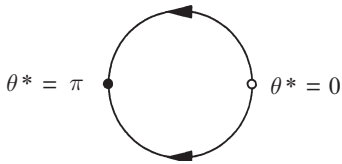


Figure 4.1.1

Actually, we’ve seen this example before—it’s given in Section 2.1. There we regarded $\dot{x} = \sin x$ as a vector field on the *line*. Compare Figure 2.1.1 with Figure 4.1.1 and notice how much clearer it is to think of this system as a vector field on the circle. ■

EXAMPLE 4.1.2:

Explain why $\dot{\theta} = \theta$ cannot be regarded as a vector field on the circle, for θ in the range $-\infty < \theta < \infty$.

Solution: The velocity is not uniquely defined. For example, $\theta = 0$ and $\theta = 2\pi$ are two labels for the same point on the circle, but the first label implies a velocity of 0 at that point, while the second implies a velocity of 2π . ■

If we try to avoid this non-uniqueness by restricting θ to the range $-\pi < \theta \leq \pi$, then the velocity vector jumps discontinuously at the point corresponding to $\theta = \pi$. Try as we might, there’s no way to consider $\dot{\theta} = \theta$ as a smooth vector field on the entire circle.

Of course, there’s no problem regarding $\dot{\theta} = \theta$ as a vector field on the *line*, because then $\theta = 0$ and $\theta = 2\pi$ are different points, and so there’s no conflict about how to define the velocity at each of them.

Example 4.1.2 suggests how to define vector fields on the circle. Here’s a geometric definition: A **vector field on the circle** is a rule that assigns a unique velocity vector to each point on the circle.

In practice, such vector fields arise when we have a first-order system $\dot{\theta} = f(\theta)$, where $f(\theta)$ is a real-valued, 2π -periodic function. That is, $f(\theta + 2\pi) = f(\theta)$ for all real θ . Moreover, we assume (as usual) that $f(\theta)$ is smooth enough to guarantee existence and uniqueness of solutions. Although this system could be regarded as a special case of a vector field on the line, it is usually clearer to think of it as a vector field on the circle (as in Example 4.1.1). This means that we don’t distinguish

between θ 's that differ by an integer multiple of 2π . Here's where the periodicity of $f(\theta)$ becomes important—it ensures that the velocity $\dot{\theta}$ is uniquely defined at each point θ on the circle, in the sense that $\dot{\theta}$ is the same, whether we call that point θ or $\theta + 2\pi$, or $\theta + 2\pi k$ for any integer k .

4.2 Uniform Oscillator

A point on a circle is often called an *angle* or a *phase*. Then the simplest oscillator of all is one in which the phase θ changes uniformly:

$$\dot{\theta} = \omega$$

where ω is a constant. The solution is

$$\theta(t) = \omega t + \theta_0,$$

which corresponds to uniform motion around the circle at an angular frequency ω . This solution is *periodic*, in the sense that $\theta(t)$ changes by 2π , and therefore returns to the same point on the circle, after a time $T = 2\pi/\omega$. We call T the *period* of the oscillation.

Notice that we have said nothing about the *amplitude* of the oscillation. There really is no amplitude variable in our system. If we had an amplitude as well as a phase variable, we'd be in a *two-dimensional* phase space; this situation is more complicated and will be discussed later in the book. (Or if you prefer, you can imagine that the oscillation occurs at some *fixed* amplitude, corresponding to the radius of our circular phase space. In any case, amplitude plays no role in the dynamics.)

EXAMPLE 4.2.1:

Two joggers, Speedy and Pokey, are running at a steady pace around a circular track. It takes Speedy T_1 seconds to run once around the track, whereas it takes Pokey $T_2 > T_1$ seconds. Of course, Speedy will periodically overtake Pokey; how long does it take for Speedy to lap Pokey once, assuming that they start together?

Solution: Let $\theta_1(t)$ be Speedy's position on the track. Then $\dot{\theta}_1 = \omega_1$ where $\omega_1 = 2\pi/T_1$. This equation says that Speedy runs at a steady pace and completes a circuit every T_1 seconds. Similarly, suppose that $\dot{\theta}_2 = \omega_2 = 2\pi/T_2$ for Pokey.

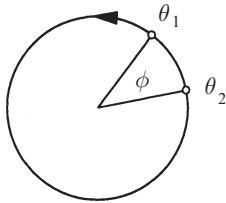


Figure 4.2.1

The condition for Speedy to lap Pokey is that the angle between them has increased by 2π . Thus if we define the *phase difference* $\phi = \theta_1 - \theta_2$, we want to find how long it takes for ϕ to increase by 2π (Figure 4.2.1). By subtraction we find $\dot{\phi} = \dot{\theta}_1 - \dot{\theta}_2 = \omega_1 - \omega_2$. Thus ϕ increases by 2π after a time

$$T_{\text{lap}} = \frac{2\pi}{\omega_1 - \omega_2} = \left(\frac{1}{T_1} - \frac{1}{T_2} \right)^{-1}. \blacksquare$$

Example 4.2.1 illustrates an effect called the **beat phenomenon**. Two noninteracting oscillators with different frequencies will periodically go in and out of phase with each other. You may have heard this effect on a Sunday morning: sometimes the bells of two different churches will ring simultaneously, then slowly drift apart, and then eventually ring together again. If the oscillators *interact* (for example, if the two joggers try to stay together or the bell ringers can hear each other), then we can get more interesting effects, as we will see in Section 4.5 on the flashing rhythm of fireflies.

4.3 Nonuniform Oscillator

The equation

$$\dot{\theta} = \omega - a \sin \theta \tag{1}$$

arises in many different branches of science and engineering. Here is a partial list:

Electronics (phase-locked loops)

Biology (oscillating neurons, firefly flashing rhythm, human sleep-wake cycle)

Condensed-matter physics (Josephson junction, charge-density waves)

Mechanics (Overdamped pendulum driven by a constant torque)

Some of these applications will be discussed later in this chapter and in the exercises.

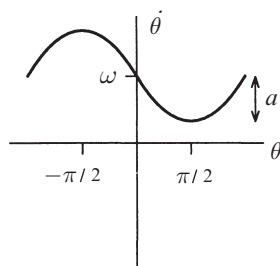


Figure 4.3.1

To analyze (1), we assume that $\omega > 0$ and $a \geq 0$ for convenience; the results for negative ω and a are similar. A typical graph of $f(\theta) = \omega - a \sin \theta$ is shown in Figure 4.3.1. Note that ω is the mean and a is the amplitude.

Vector Fields

If $a = 0$, (1) reduces to the uniform oscillator. The parameter a introduces a *nonuniformity* in the flow around the circle: the flow is fastest at $\theta = -\pi/2$ and slowest at $\theta = \pi/2$ (Figure 4.3.2a). This nonuniformity becomes more pronounced as a increases. When a is slightly less than ω , the oscillation is very jerky: the phase point $\theta(t)$ takes a long time to pass through a **bottleneck** near $\theta = \pi/2$, after which it zips around the rest of the circle on a much faster time scale. When $a = \omega$, the system stops oscillating

altogether: a half-stable fixed point has been born in a *saddle-node bifurcation* at $\theta = \pi/2$ (Figure 4.3.2b). Finally, when $a > \omega$, the half-stable fixed point splits into a stable and unstable fixed point (Figure 4.3.2c). All trajectories are attracted to the stable fixed point as $t \rightarrow \infty$.

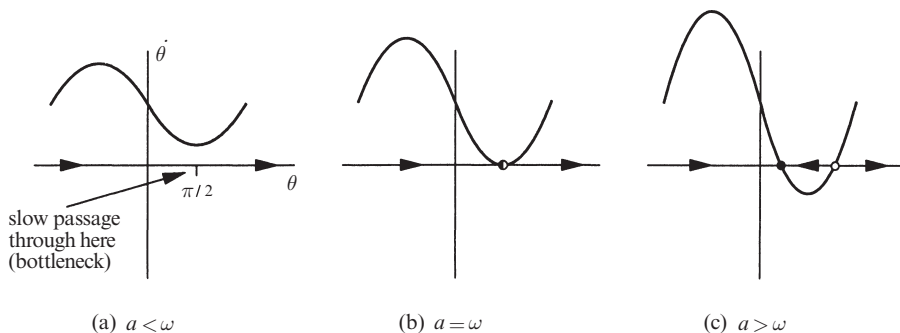


Figure 4.3.2

The same information can be shown by plotting the vector fields on the circle (Figure 4.3.3).

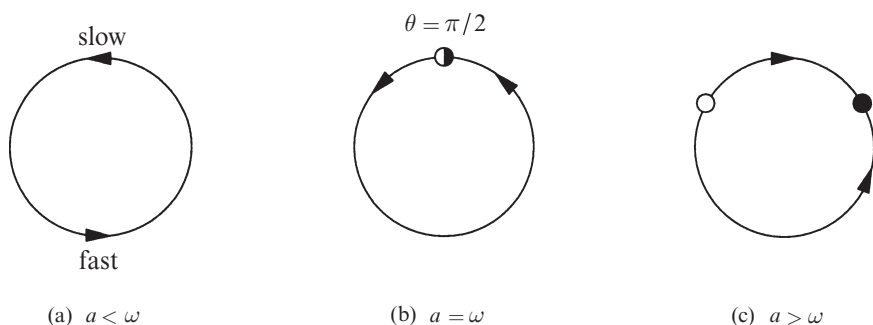


Figure 4.3.3

EXAMPLE 4.3.1:

Use linear stability analysis to classify the fixed points of (1) for $a > \omega$.

Solution: The fixed points θ^* satisfy

$$\sin \theta^* = \omega/a, \quad \cos \theta^* = \pm \sqrt{1 - (\omega/a)^2}.$$

Their linear stability is determined by

$$f'(\theta^*) = -a \cos \theta^* = \mp a \sqrt{1 - (\omega/a)^2}.$$

Thus the fixed point with $\cos \theta^* > 0$ is the stable one, since $f'(\theta^*) < 0$. This agrees with Figure 4.3.2c. ■

Oscillation Period

For $a < \omega$, the period of the oscillation can be found analytically, as follows: the time required for θ to change by 2π is given by

$$\begin{aligned} T &= \int dt = \int_0^{2\pi} \frac{dt}{d\theta} d\theta \\ &= \int_0^{2\pi} \frac{d\theta}{\omega - a \sin \theta} \end{aligned}$$

where we have used (1) to replace $dt / d\theta$. This integral can be evaluated by complex variable methods, or by the substitution $u = \tan \frac{\theta}{2}$. (See Exercise 4.3.2 for details.) The result is

$$T = \frac{2\pi}{\sqrt{\omega^2 - a^2}}. \quad (2)$$

Figure 4.3.4 shows the graph of T as a function of a .

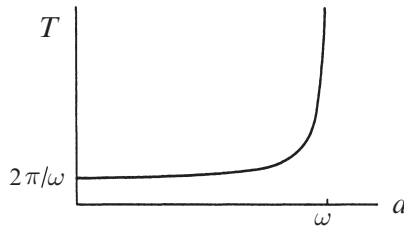


Figure 4.3.4

When $a = 0$, Equation (2) reduces to $T = 2\pi/\omega$, the familiar result for a uniform oscillator. The period increases with a and diverges as a approaches ω from below (we denote this limit by $a \rightarrow \omega^-$).

We can estimate the order of the divergence by noting that

$$\begin{aligned} \sqrt{\omega^2 - a^2} &= \sqrt{\omega + a} \sqrt{\omega - a} \\ &\approx \sqrt{2\omega} \sqrt{\omega - a} \end{aligned}$$

as $a \rightarrow \omega^-$. Hence

$$T \approx \left(\frac{\pi\sqrt{2}}{\sqrt{\omega}} \right) \frac{1}{\sqrt{\omega - a}}, \quad (3)$$

which shows that T blows up like $(a_c - a)^{-1/2}$, where $a_c = \omega$. Now let's explain the origin of this *square-root scaling law*.

Ghosts and Bottlenecks

The square-root scaling law found above is a *very general feature of systems that are close to a saddle-node bifurcation*. Just after the fixed points collide, there is a saddle-node remnant or *ghost* that leads to slow passage through a bottleneck.

For example, consider $\dot{\theta} = \omega - a \sin \theta$ for decreasing values of a , starting with $a > \omega$. As a decreases, the two fixed points approach each other, collide, and disappear (this sequence was shown earlier in Figure 4.3.3, except now you have to read from right to left.) For a slightly less than ω , the fixed points near $\pi/2$ no longer exist, but they still make themselves felt through a saddle-node ghost (Figure 4.3.5).

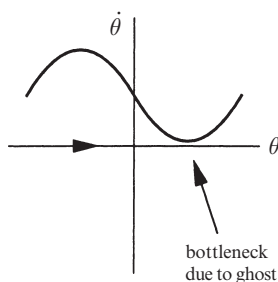


Figure 4.3.5

A graph of $\theta(t)$ would have the shape shown in Figure 4.3.6. Notice how the trajectory spends practically all its time getting through the bottleneck.

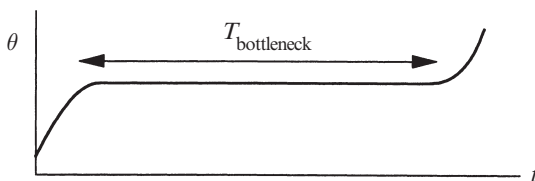


Figure 4.3.6

Now we want to derive a general scaling law for the time required to pass through a bottleneck. The only thing that matters is the behavior of $\dot{\theta}$ in the immediate vicinity of the minimum, since the time spent there dominates all other time scales in the problem. Generically, $\dot{\theta}$ looks *parabolic* near its minimum. Then the problem simplifies tremendously: the dynamics can be reduced to the normal

form for a saddle-node bifurcation! By a local rescaling of space, we can rewrite the vector field as

$$\dot{x} = r + x^2$$

where r is proportional to the distance from the bifurcation, and $0 < r \ll 1$. The graph of $\dot{x}$ is shown in Figure 4.3.7.

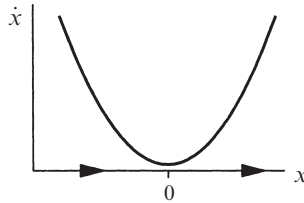


Figure 4.3.7

To estimate the time spent in the bottleneck, we calculate the time taken for x to go from $-\infty$ (all the way on one side of the bottleneck) to $+\infty$ (all the way on the other side). The result is

$$T_{\text{bottleneck}} \approx \int_{-\infty}^{\infty} \frac{dx}{r + x^2} = \frac{\pi}{\sqrt{r}}, \quad (4)$$

which shows the generality of the square-root scaling law. (Exercise 4.3.1 reminds you how to evaluate the integral in (4).)

EXAMPLE 4.3.2:

Estimate the period of $\dot{\theta} = \omega - a \sin \theta$ in the limit $a \rightarrow \omega^-$, using the normal form method instead of the exact result.

Solution: The period will be essentially the time required to get through the bottleneck. To estimate this time, we use a Taylor expansion about $\theta = \pi/2$, where the bottleneck occurs. Let $\phi = \theta - \pi/2$, where ϕ is small. Then

$$\begin{aligned} \dot{\phi} &= \omega - a \sin(\phi + \frac{\pi}{2}) \\ &= \omega - a \cos \phi \\ &= \omega - a + \frac{1}{2} a \phi^2 + \cdots \end{aligned}$$

which is close to the desired normal form. If we let

$$x = (a/2)^{1/2} \phi, \quad r = \omega - a$$

then $(2/a)^{1/2} \dot{x} \approx r + x^2$, to leading order in x . Separating variables yields

$$T \approx (2/a)^{1/2} \int_{-\infty}^{\infty} \frac{dx}{r+x^2} = (2/a)^{1/2} \frac{\pi}{\sqrt{r}}.$$

Now we substitute $r = \omega - a$. Furthermore, since $a \rightarrow \omega^-$, we may replace $2/a$ by $2/\omega$. Hence

$$T \approx \left(\frac{\pi\sqrt{2}}{\sqrt{\omega}} \right) \frac{1}{\sqrt{\omega - a}},$$

which agrees with (3). ■

4.4 Overdamped Pendulum

We now consider a simple mechanical example of a nonuniform oscillator: an overdamped pendulum driven by a constant torque. Let θ denote the angle between the pendulum and the downward vertical, and suppose that θ increases counterclockwise (Figure 4.4.1).

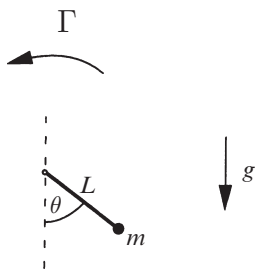


Figure 4.4.1

Then Newton's law yields

$$mL^2\ddot{\theta} + b\dot{\theta} + mgL\sin\theta = \Gamma \quad (1)$$

where m is the mass and L is the length of the pendulum, b is a viscous damping constant, g is the acceleration due to gravity, and Γ is a constant applied torque. All of these parameters are positive. In particular, $\Gamma > 0$ implies that the applied torque drives the pendulum counterclockwise, as shown in Figure 4.4.1.

Equation (1) is a second-order system, but in the *overdamped limit* of extremely large b , it may be approximated by a first-order system (see Section 3.5 and Exercise 4.4.1). In this limit the inertia term $mL^2\ddot{\theta}$ is negligible and so (1) becomes

$$b\dot{\theta} + mgL\sin\theta = \Gamma. \quad (2)$$

To think about this problem physically, you should imagine that the pendulum is immersed in molasses. The torque Γ enables the pendulum to plow through its viscous surroundings. Please realize that this is the *opposite* limit from the familiar frictionless case in which energy is conserved, and the pendulum swings back and forth forever. In the present case, energy is lost to damping and pumped in by the applied torque.

To analyze (2), we first nondimensionalize it. Dividing by mgL yields

$$\frac{b}{mgL} \dot{\theta} = \frac{\Gamma}{mgL} - \sin \theta.$$

Hence, if we let

$$\tau = \frac{mgL}{b} t, \quad \gamma = \frac{\Gamma}{mgL} \quad (3)$$

then

$$\theta' = \gamma - \sin \theta \quad (4)$$

where $\theta' = d\theta/d\tau$.

The dimensionless group γ is the ratio of the applied torque to the maximum gravitational torque. If $\gamma > 1$ then the applied torque can never be balanced by the gravitational torque and *the pendulum will overturn continually*. The rotation rate is nonuniform, since gravity helps the applied torque on one side and opposes it on the other (Figure 4.4.2).

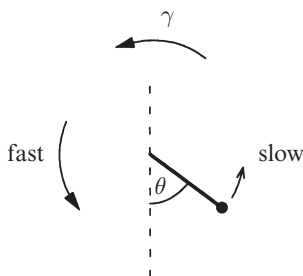


Figure 4.4.2

As $\gamma \rightarrow 1^+$, the pendulum takes longer and longer to climb past $\theta = \pi/2$ on the slow side. When $\gamma = 1$ a fixed point appears at $\theta^* = \pi/2$, and then splits into two when $\gamma < 1$ (Figure 4.4.3). On physical grounds, it's clear that the lower of the two equilibrium positions is the stable one.

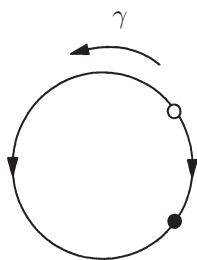


Figure 4.4.3

As γ decreases, the two fixed points move farther apart. Finally, when $\gamma = 0$, the applied torque vanishes and there is an unstable equilibrium at the top (inverted pendulum) and a stable equilibrium at the bottom.

4.5 Fireflies

Fireflies provide one of the most spectacular examples of synchronization in nature. In some parts of southeast Asia, thousands of male fireflies gather in trees at night and flash on and off in unison. Meanwhile the female fireflies cruise overhead, looking for males with a handsome light.

To really appreciate this amazing display, you have to see a movie or videotape of it. A good example is shown in David Attenborough's (1992) television series *The Trials of Life*, in the episode called "Talking to Strangers." See Buck and Buck (1976) for a beautifully written introduction to synchronous fireflies, and Buck (1988) for a comprehensive review. For mathematical models of synchronous fireflies, see Mirollo and Strogatz (1990) and Ermentrout (1991).

How does the synchrony occur? Certainly the fireflies don't start out synchronized; they arrive in the trees at dusk, and the synchrony builds up gradually as the night goes on. The key is that *the fireflies influence each other*: When one firefly sees the flash of another, it slows down or speeds up so as to flash more nearly in phase on the next cycle.

Hanson (1978) studied this effect experimentally, by periodically flashing a light at a firefly and watching it try to synchronize. For a range of periods close to the firefly's natural period (about 0.9 sec), the firefly was able to match its frequency to the periodic stimulus. In this case, one says that the firefly had been *entrained* by the stimulus. However, if the stimulus was too fast or too slow, the firefly could not keep up and entrainment was lost—then a kind of beat phenomenon occurred. But in contrast to the simple beat phenomenon of Section 4.2, the phase difference between stimulus and firefly did not increase uniformly. The phase difference increased slowly during part of the beat cycle, as the firefly struggled in vain to synchronize, and then it increased rapidly through 2π , after which the firefly tried again on the next beat cycle. This process is called *phase walkthrough* or *phase drift*.